Intro2CS OOH

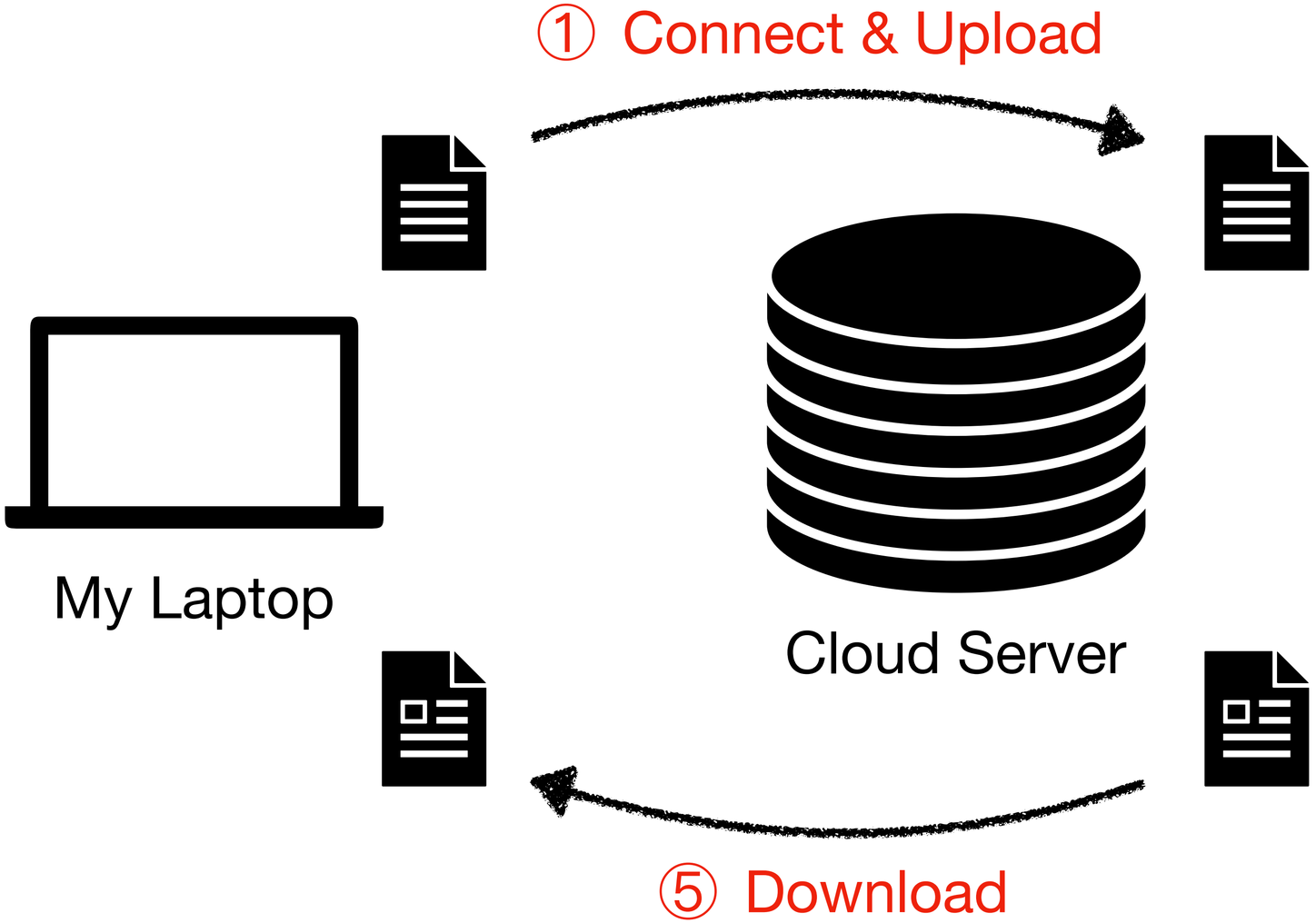
How to Use a Cloud Server?

Jinchen Jiang

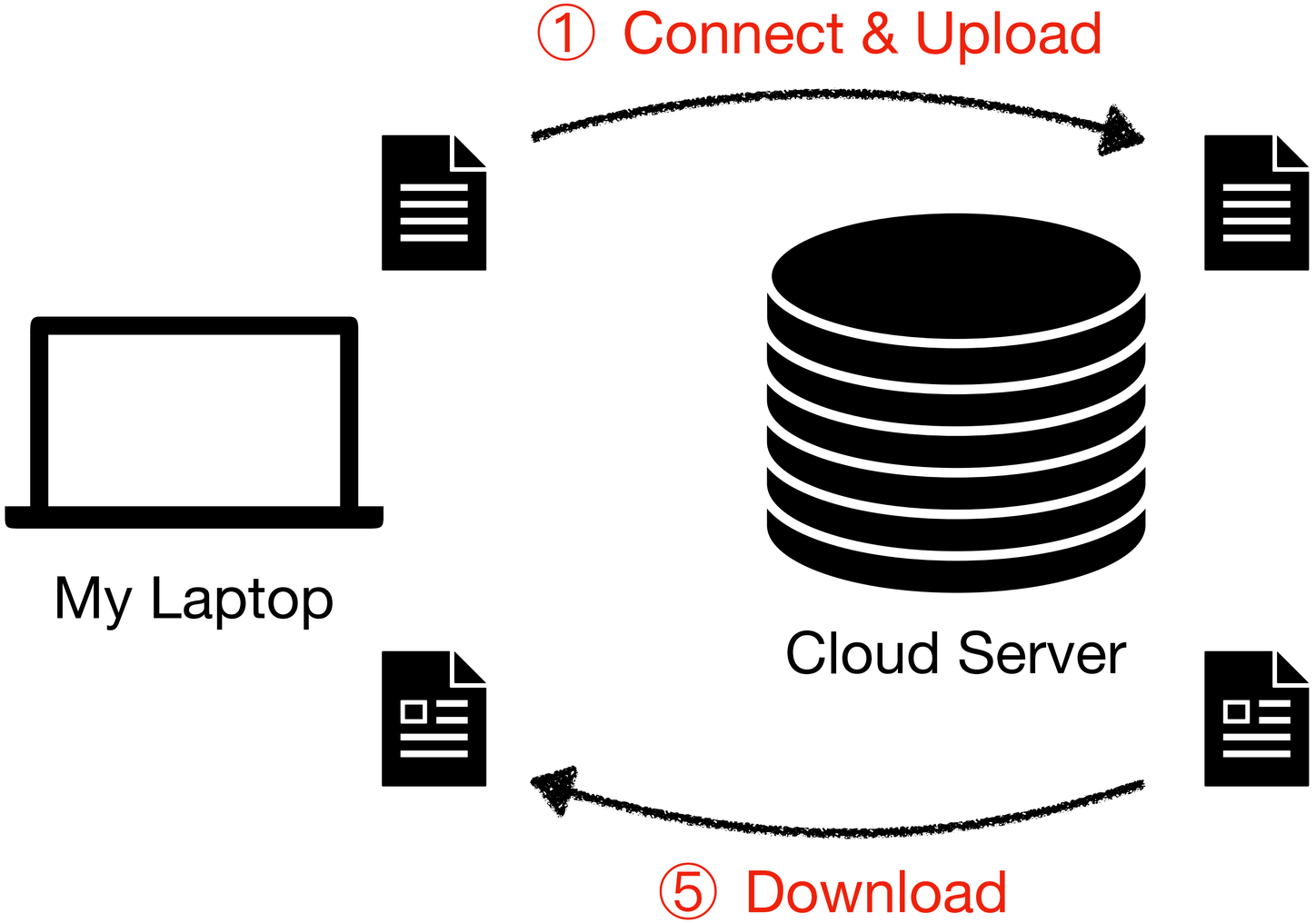
Why a Cloud Server?

Laptop	Server
8 CPU cores	64+ CPU cores
16 GB RAM	512+ GB RAM
Maybe no GPU	Multiple GPUs
Sleeps when you close it	Runs 24/7
Mostly yours	Shared

Today’s Mission



Today’s Mission

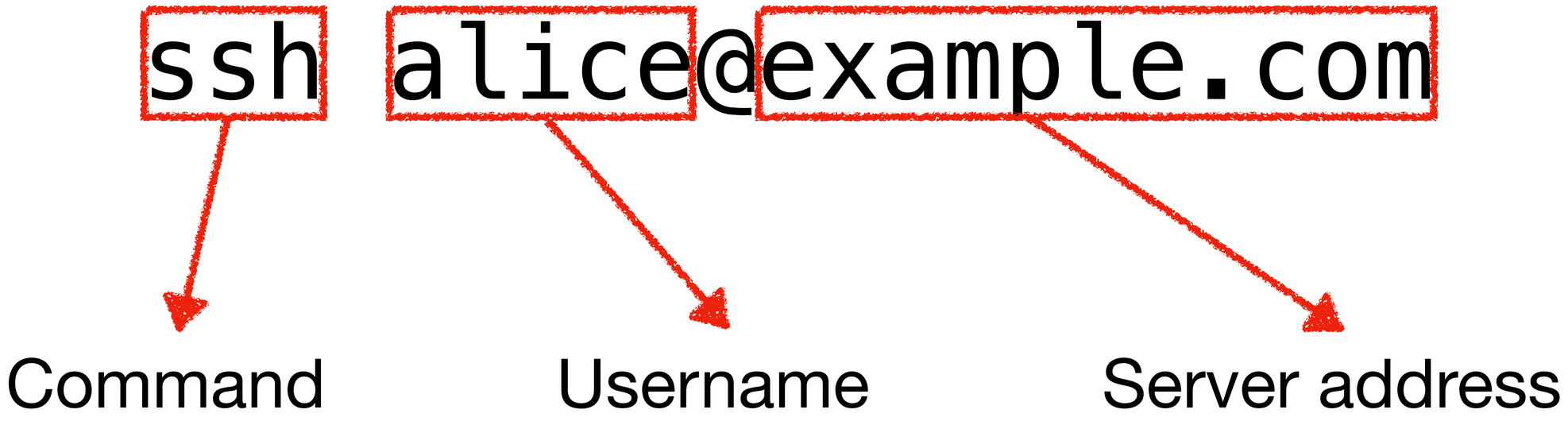


② Run it

③ Leave it alive

④ Don’t disturb others

SSH: Get into Your Server



But How Does the Server Know It’s You?

- Basically, two common approaches:

① Password

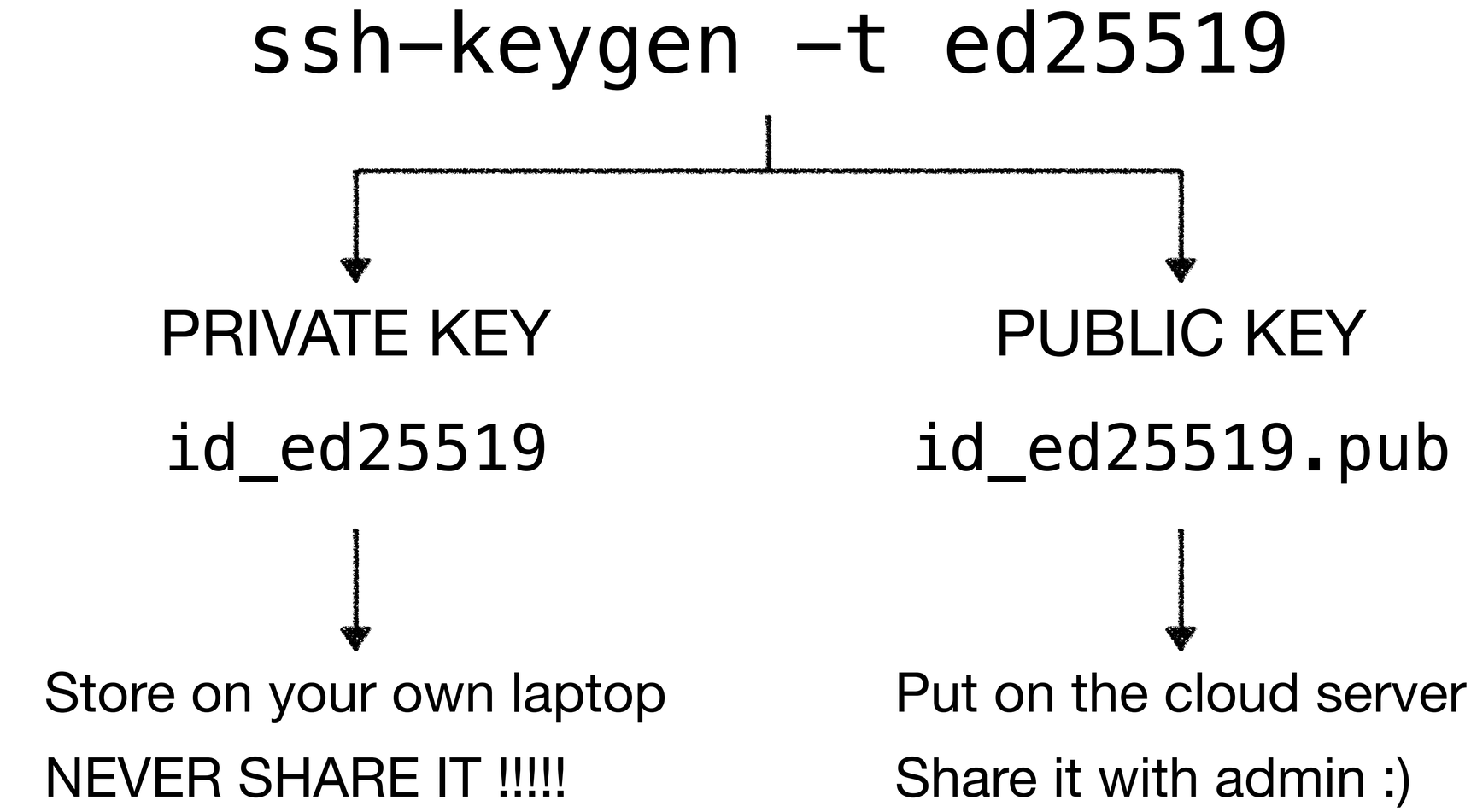
Something you know. Just use it.

② SSH key

Cryptographic identity.

- For servers you use regularly, SSH keys are extremely common.

SSH key pair: One Public, One Private.



LIVE: Let’s Connect!

Two computers, two file systems

- How can we transfer files between your laptop and the server?

→ SCP helps!

  - Upload

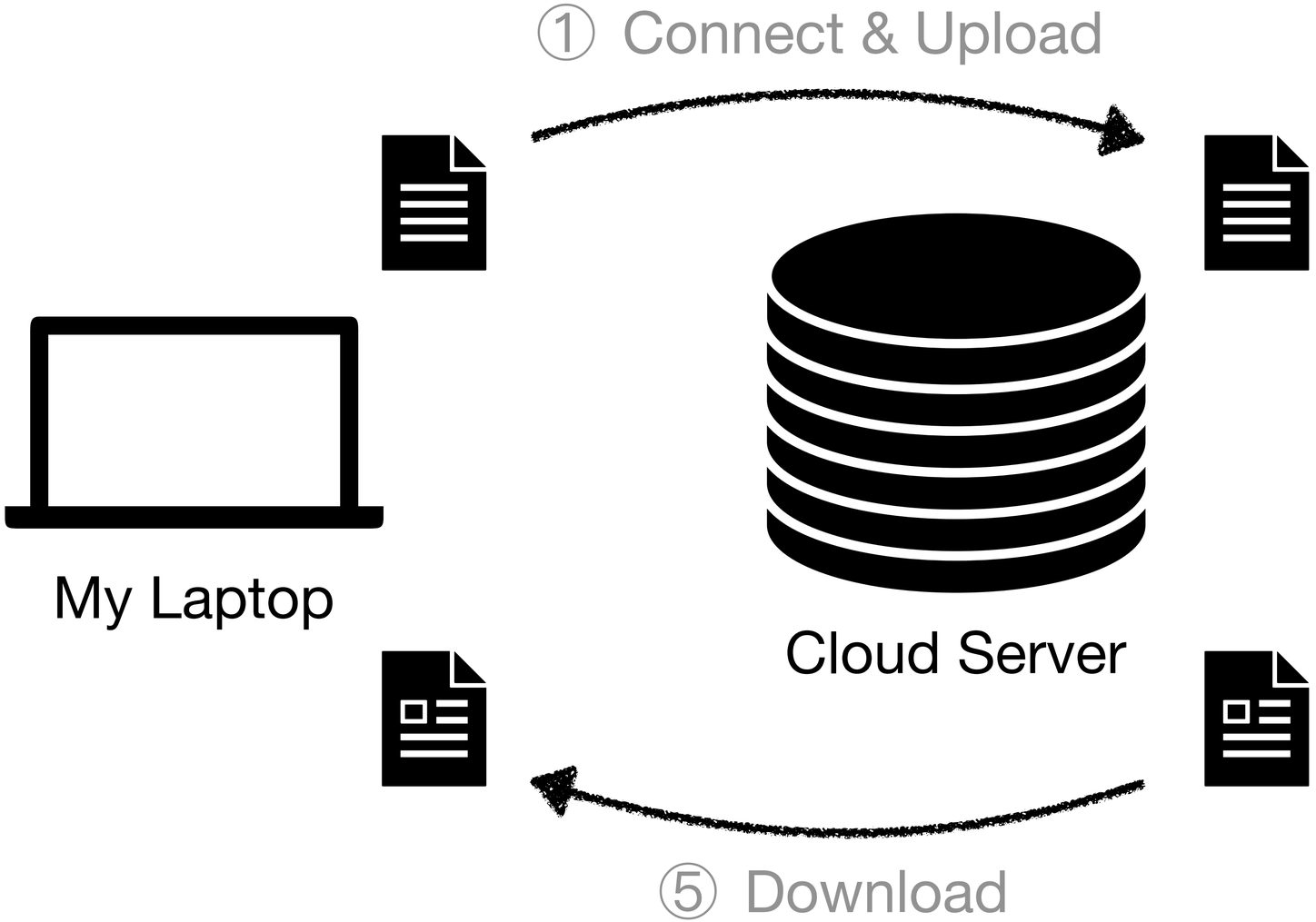
```
scp work.py alice@server:~/
```

  - Download

```
scp alice@server:~/result.txt .
```

- Run these commands on your laptop :)

Today’s Mission

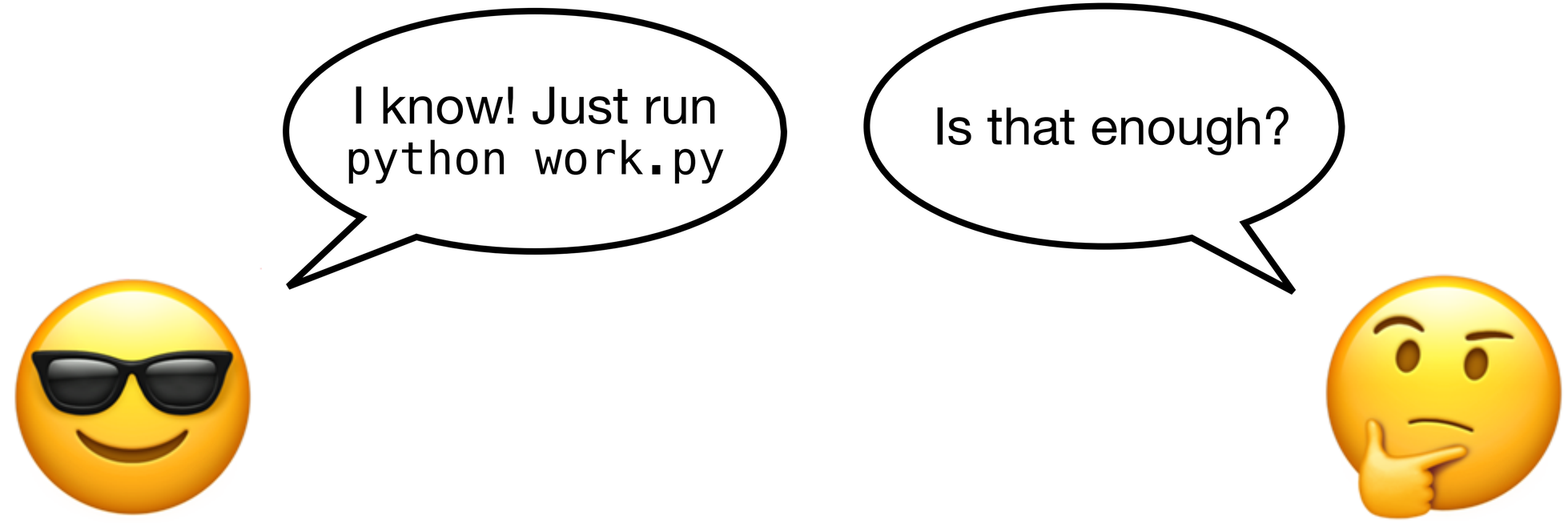


② Run it

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Run a Program

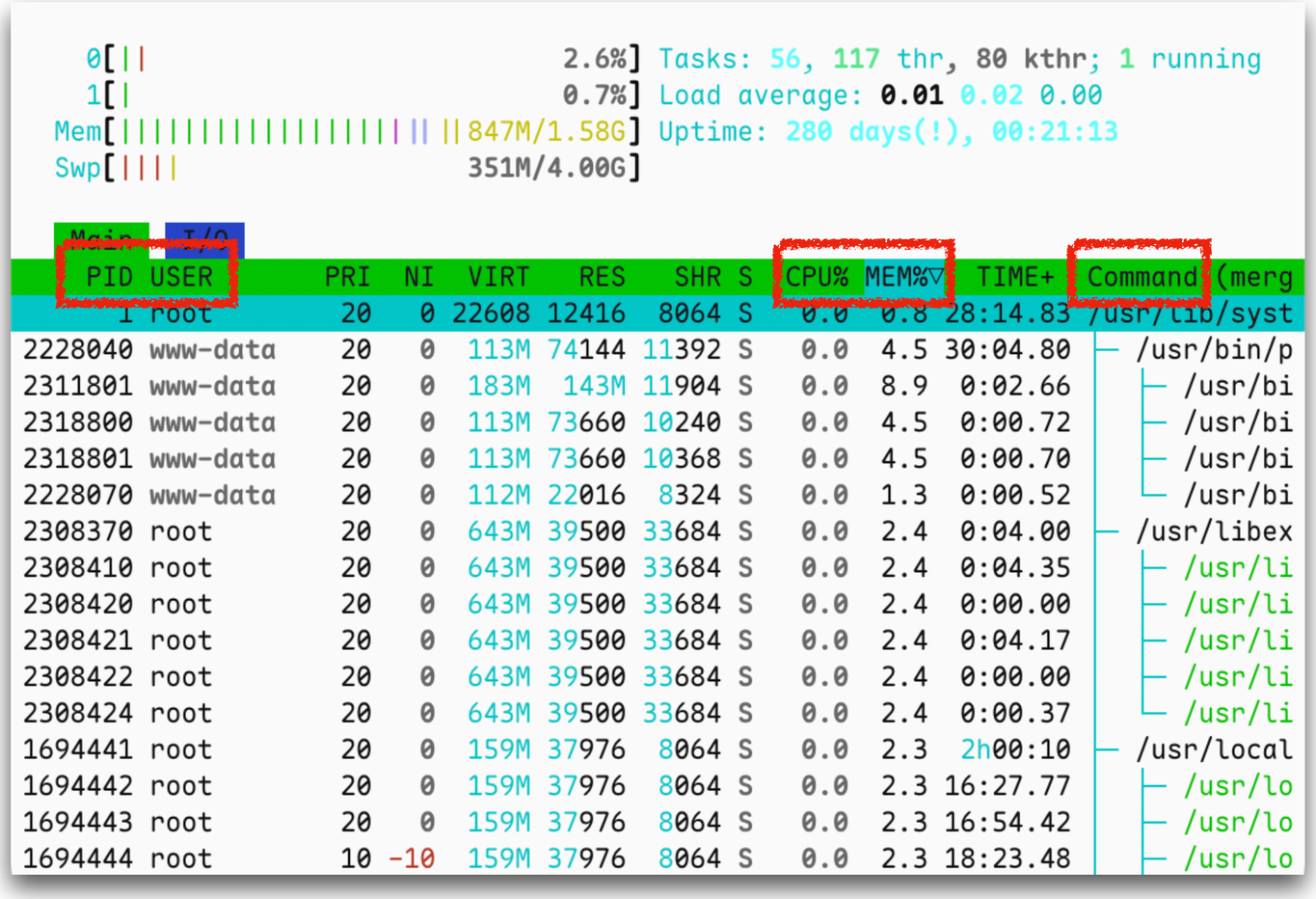


A Server is Usually Shared

- Before running something expensive, understand what resources are already being used.
- Some useful commands:

`htop`	processes / CPU / RAM
`free -h`	memory
`df -h`	disk
`w`	logged-in users

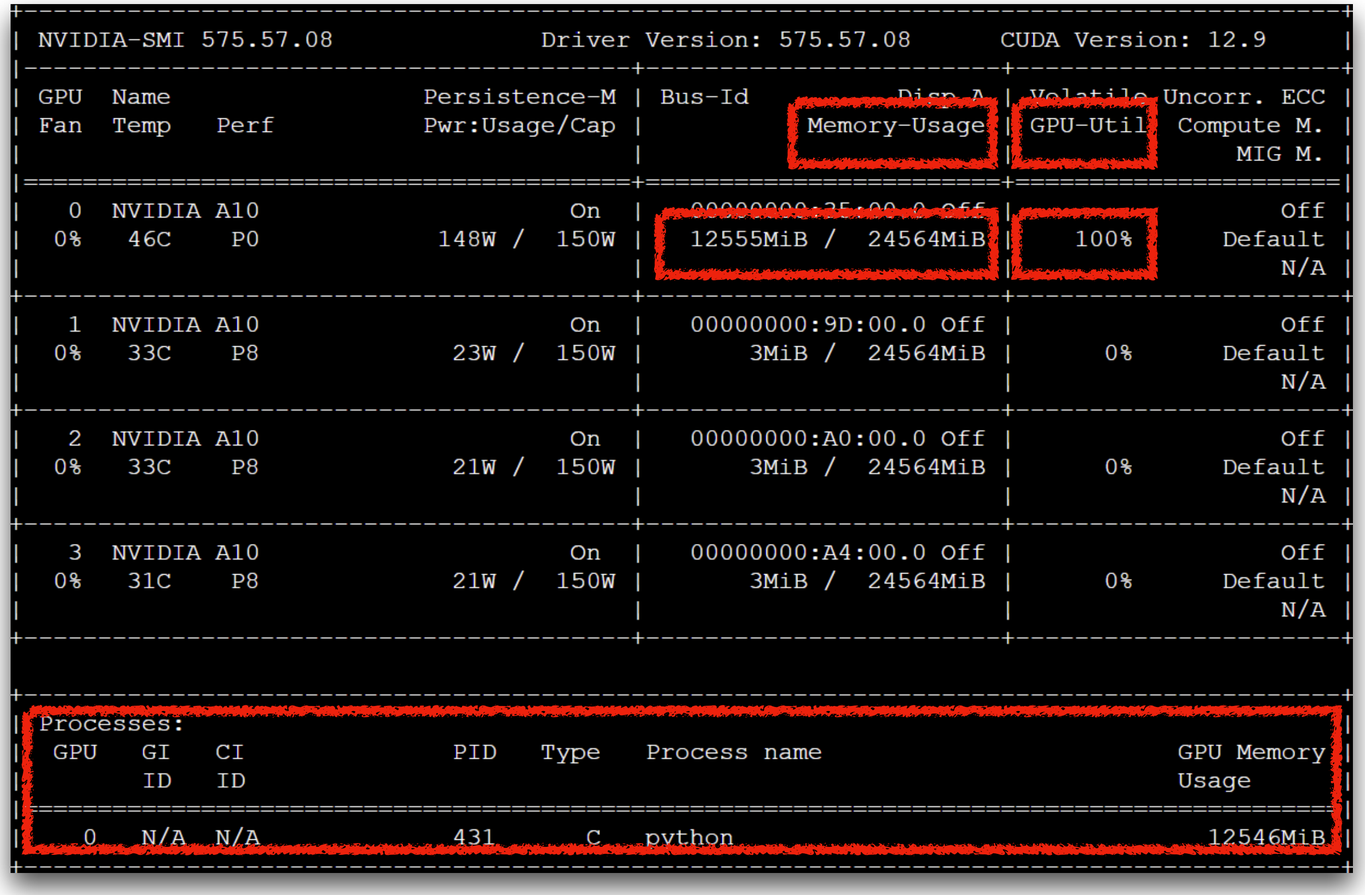
htop: What's Happening?



Answer 4 questions:

1. Who?
2. Which process?
3. How much CPU%?
4. How much MEM%?

nvidia-smi: Who’s Using GPUs?



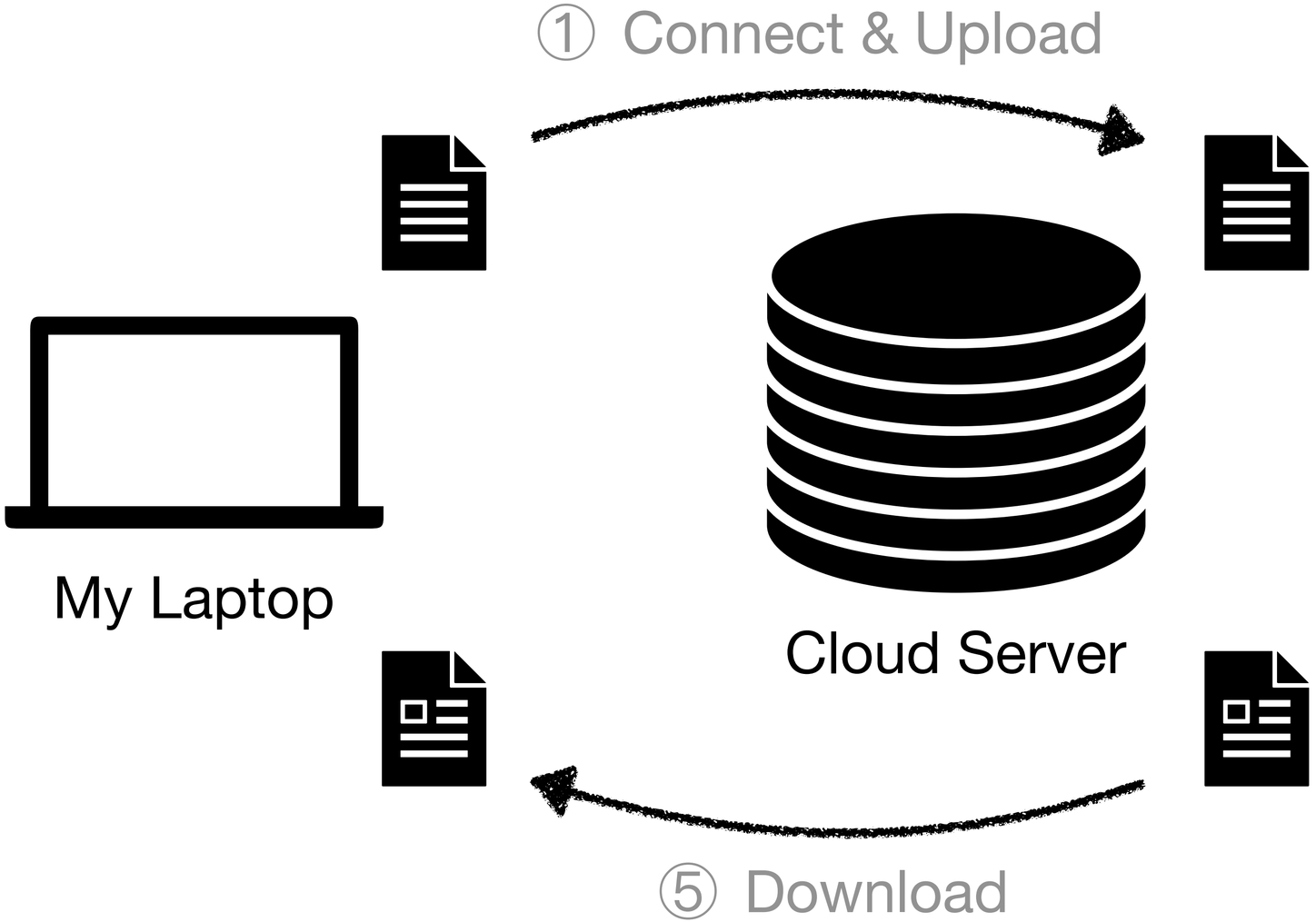
```
+-----------------------------------------------------------------------------------------+
| NVIDIA-SMI 575.57.08              Driver Version: 575.57.08      CUDA Version: 12.9     |
|-----------------------------------------+------------------------+----------------------+
| GPU  Name                 Persistence-M | Bus-Id          Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |           Memory-Usage | GPU-Util  Compute M. |
|                                         |                        |               MIG M. |
|=========================================+========================+======================|
|   0  NVIDIA A10                     On  |   00000000:35:00.0 Off |                    Off |
|  0%   46C    P0            148W /  150W |   12555MiB /  24564MiB |     100%      Default |
|                                         |                        |                  N/A |
+-----------------------------------------+------------------------+----------------------+
|   1  NVIDIA A10                     On  |   00000000:9D:00.0 Off |                    Off |
|  0%   33C    P8             23W /  150W |       3MiB /  24564MiB |      0%       Default |
|                                         |                        |                  N/A |
+-----------------------------------------+------------------------+----------------------+
|   2  NVIDIA A10                     On  |   00000000:A0:00.0 Off |                    Off |
|  0%   33C    P8             21W /  150W |       3MiB /  24564MiB |      0%       Default |
|                                         |                        |                  N/A |
+-----------------------------------------+------------------------+----------------------+
|   3  NVIDIA A10                     On  |   00000000:A4:00.0 Off |                    Off |
|  0%   31C    P8             21W /  150W |       3MiB /  24564MiB |      0%       Default |
|                                         |                        |                  N/A |
+-----------------------------------------+------------------------+----------------------+

+-----------------------------------------------------------------------------------------+
| Processes:                                                                              |
|  GPU   GI   CI              PID   Type   Process name                      GPU Memory   |
|        ID   ID                                                             Usage        |
|=========================================================================================|
|    0   N/A  N/A            431      C   python                                12546MiB |
+-----------------------------------------------------------------------------------------+
```

Check 3 points:

1. Memory-Usage
2. GPU-Util
3. Processes

Today’s Mission



② Run it

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My Job takes over 24 hours…

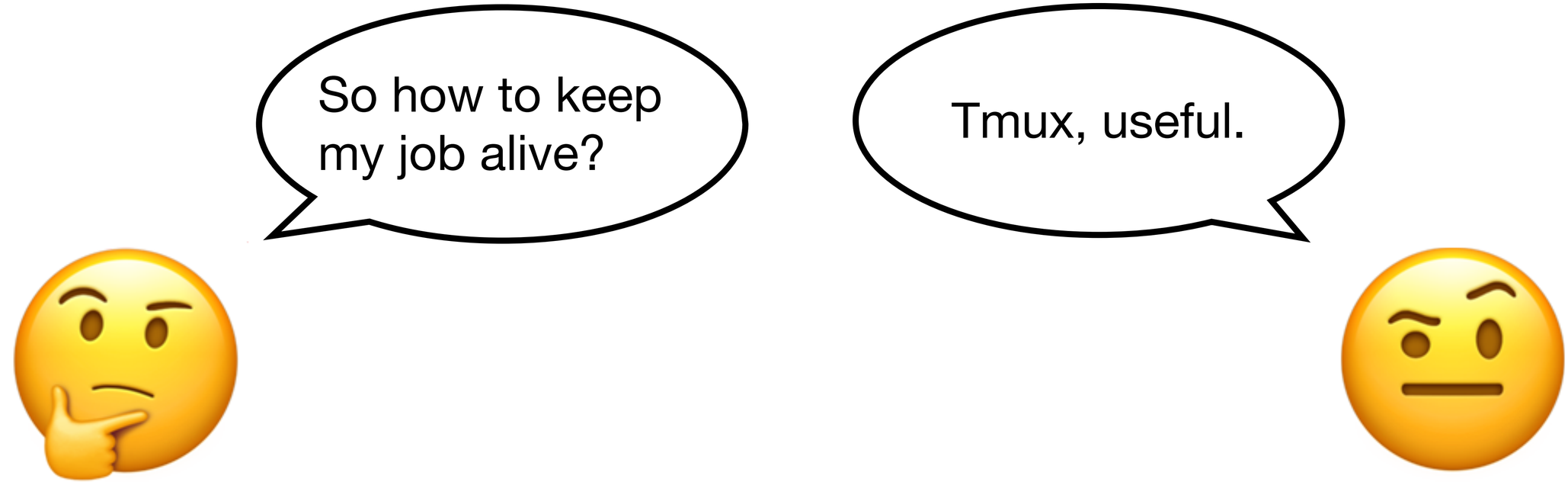


If I close my laptop, what will happen?

Connection, down. Processes, killed.



My Job takes over 24 hours…



`tmux`: Keep Your Terminal Session Alive

- Start

```
tmux new -s session_name
```

- Detach

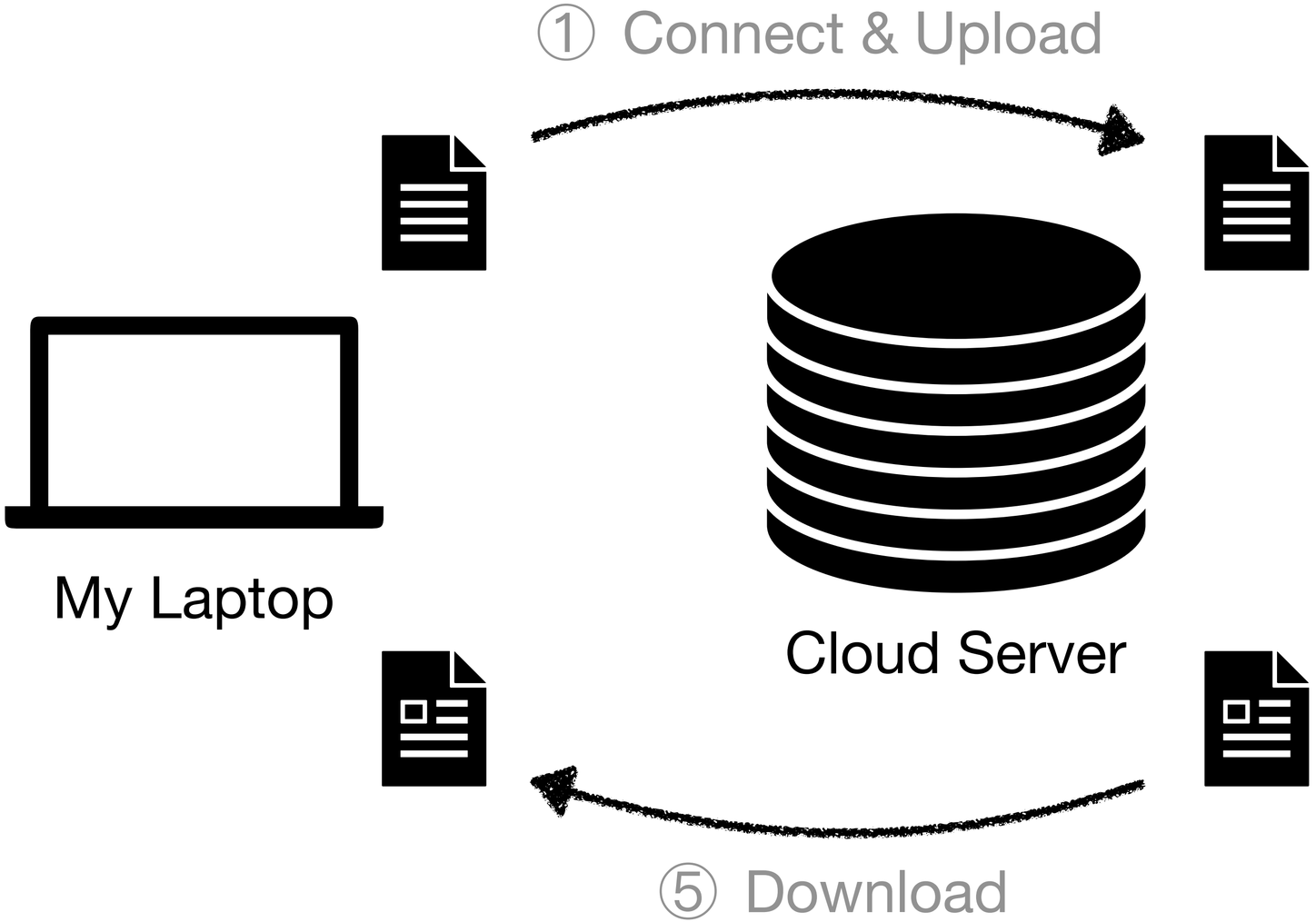
```
Ctrl+B, then D
```

- Return

```
tmux a -t session_name
```

LIVE: Let’s Have a Try!

Today’s Mission



Shared Server Survival Rules

- Check resources before you run

```
htop
```

```
nvidia-smi
```

- Don’t kill other people’s jobs
- Don’t fill the disk

```
df -h
```

Shared Server Survival Rules (Cont’d)

- Be very careful with

```
rm -rf [something]
```

- Never share

private SSH keys

passwords

API tokens

Takeaways

- Connect:

  SSH, SSH keys

- File transfer:

  SCP

- Server resources:

  `htop`, `nvidia-smi`, `df -h`, ...

- Keep job alive:

  `tmux`

- Shared server survival rules